

ETF Data Analysis

YING WEN

- ## DATASET DESCRIPTION

ETF is:

a good way of doing investment, it considers to be easy, cheap, low risk.

My Goal:

how to select a good ETF

From:

Kaggle dataset

Dataset Infor:

Original file contains: **2,353 instances, 104 features**

Target Variable: Based on value of feature 'fund_return_1year' 'create my target variable '**Is profitable**'

while 'fund_return_1year' > 0 , '**Is profitable = 1**'

while 'fund_return_1year' <= 0 , '**Is profitable = 0**'

Data Clean:

remove missing data:

remove columns which data missing exceed 10%

remove rows where there have missing data

Removed identical column: fund_name, fund_extened_name

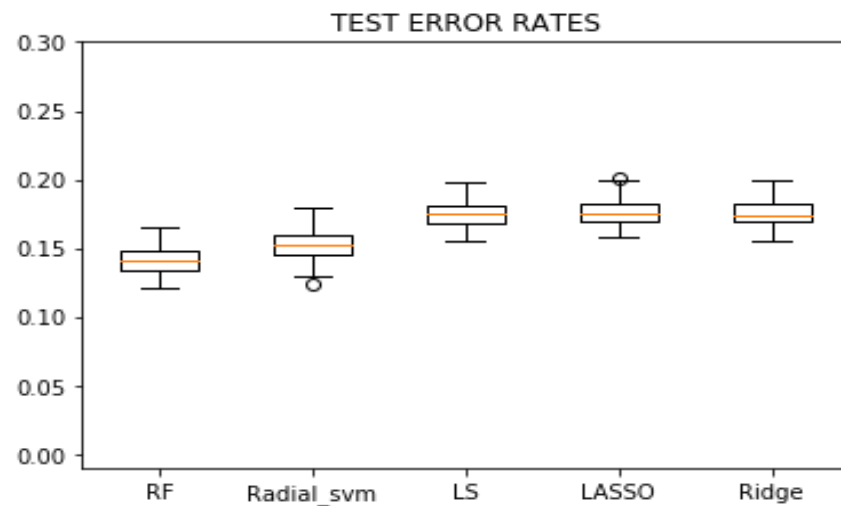
Cleaned dataset

1811 instances, 30 features, 1 Target

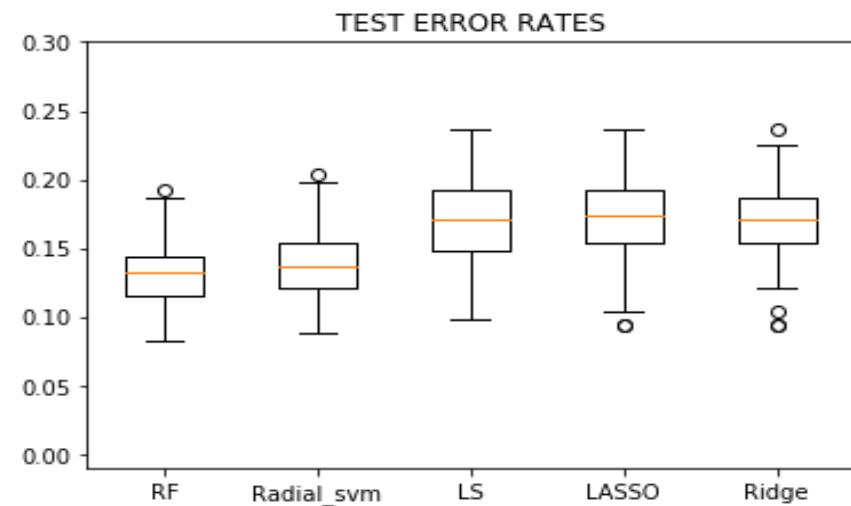
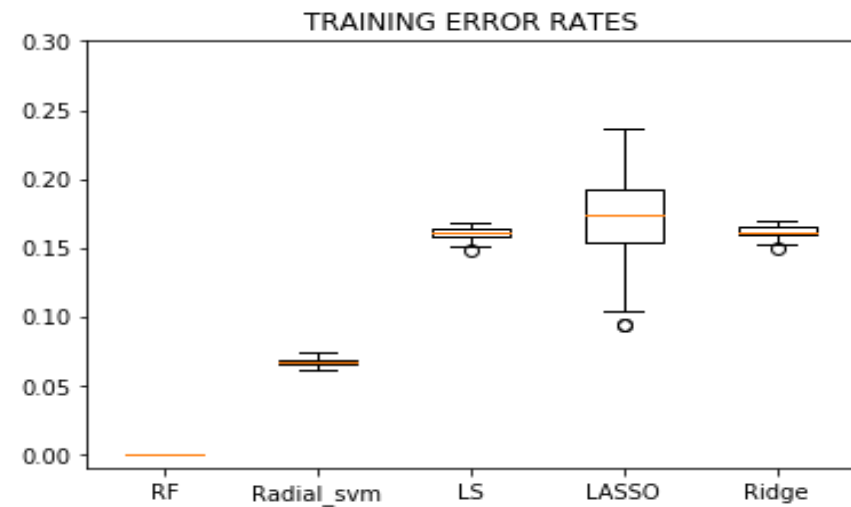
Balanced dataset

- ERROR RATES

n-learn = 0.5

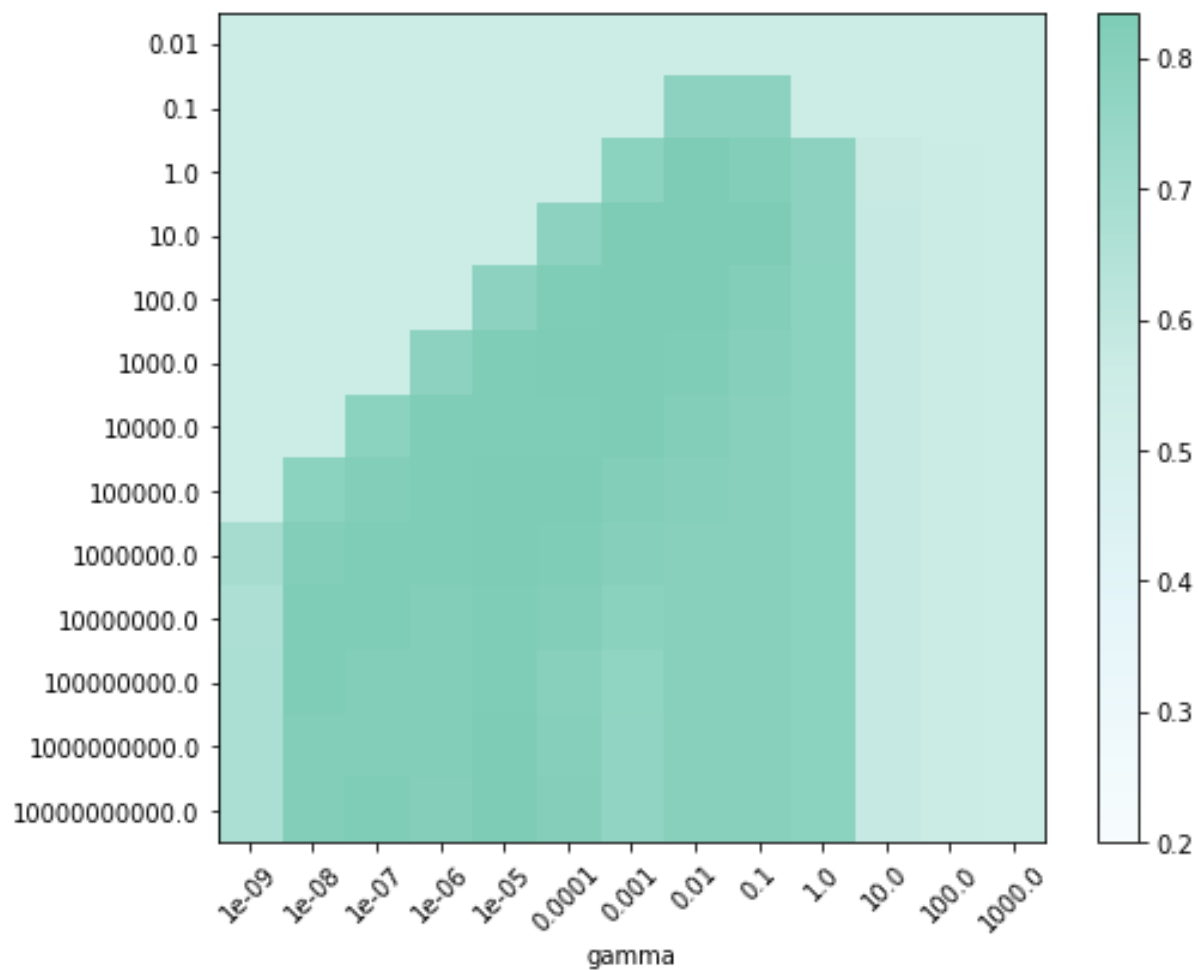


n-learn = 0.9



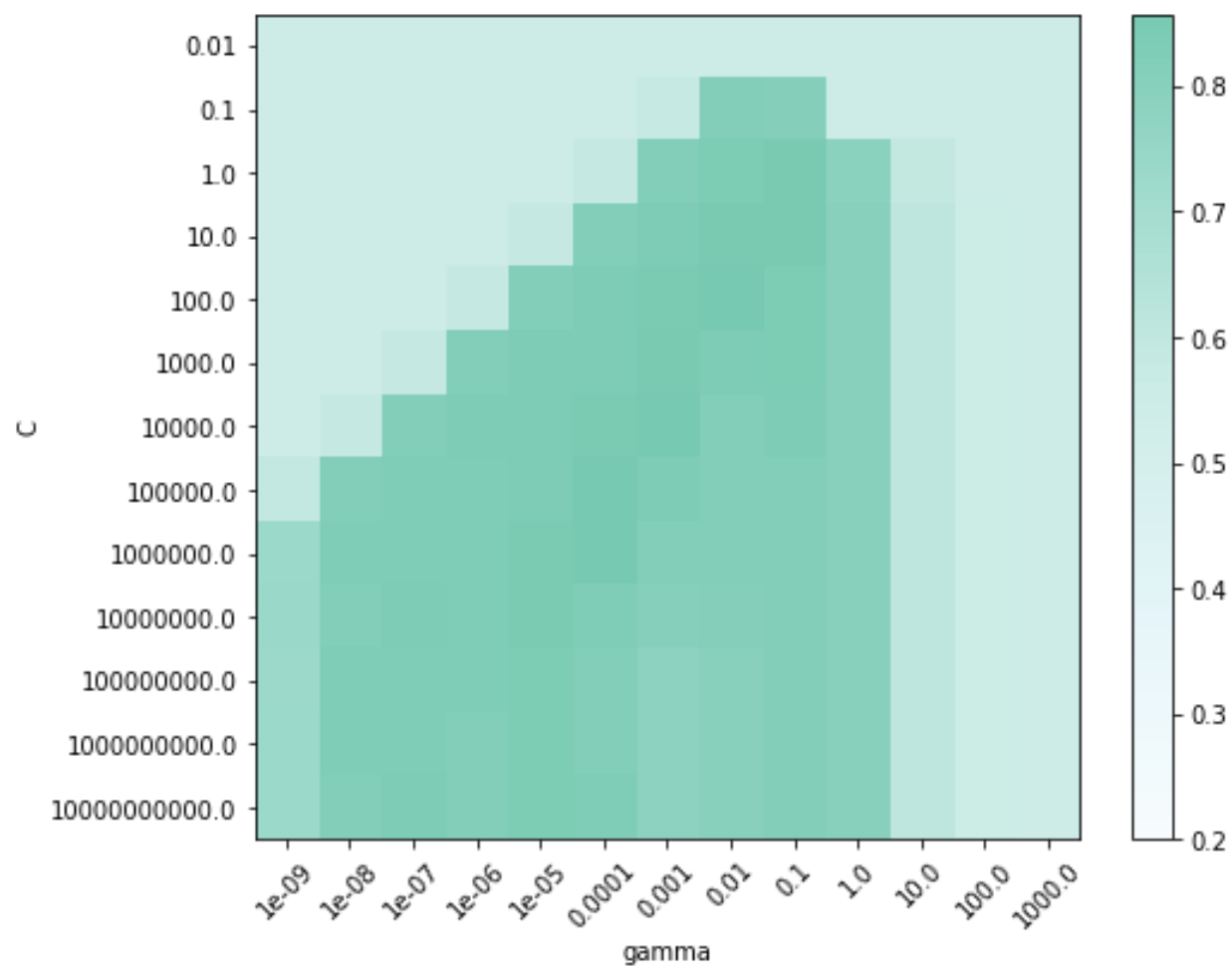
- TUNE RADIO SVM

n-learn = 0.5



The best parameters are $\{C: 1000.0, \gamma: 0.0001\}$ with a score of 0.84

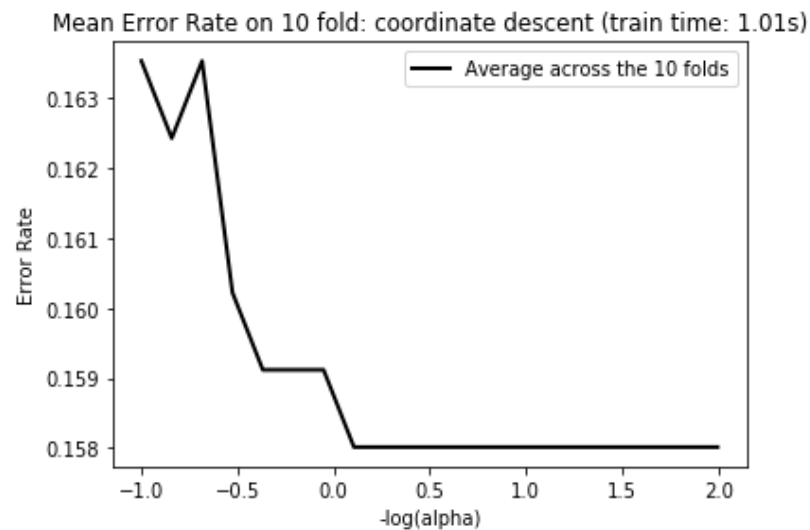
n-learn = 0.9



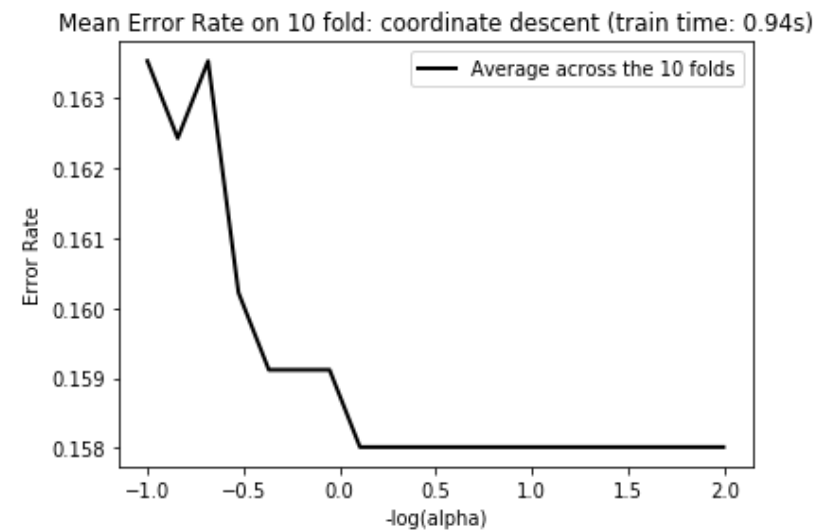
The best parameters are $\{C: 100.0, \gamma: 0.01\}$ with a score of 0.86

n-learn = 0.5

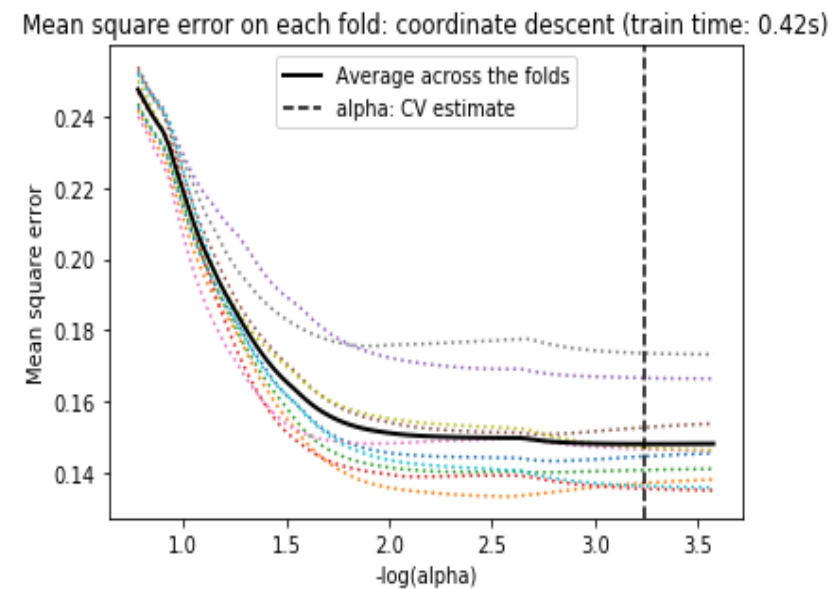
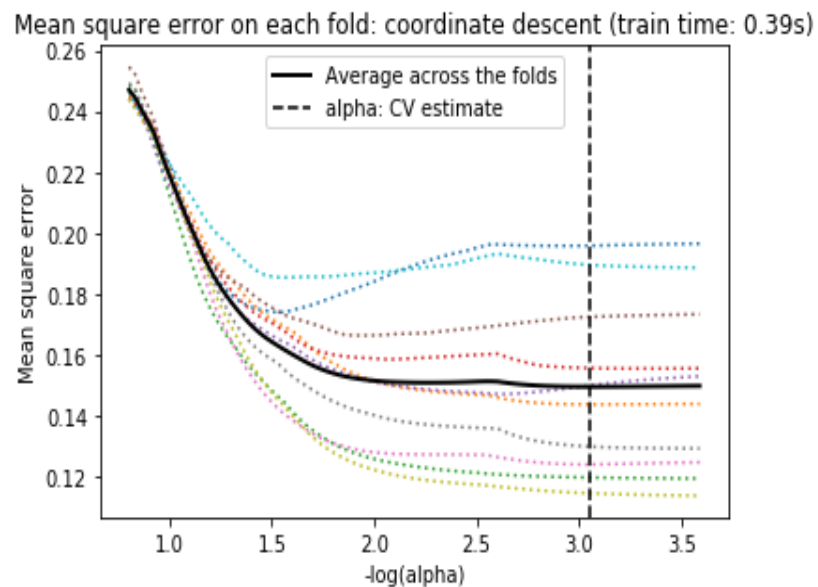
- TUNE RIDGE



n-learn = 0.9



- TUNE LASSO

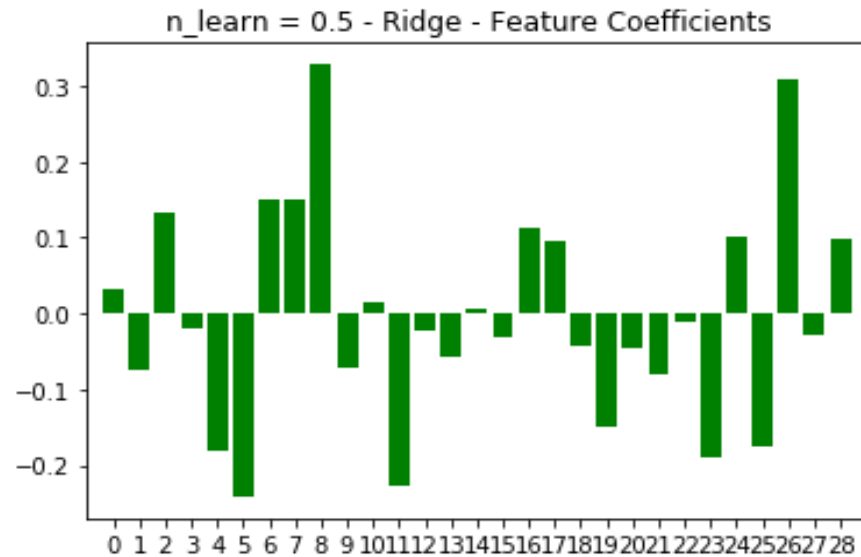
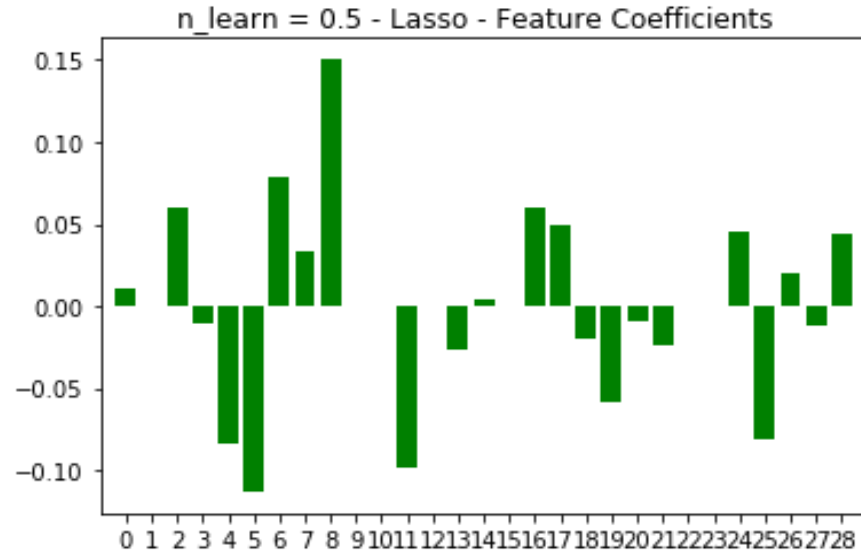


- TIME & PERFORMANCE TRADE-OFF

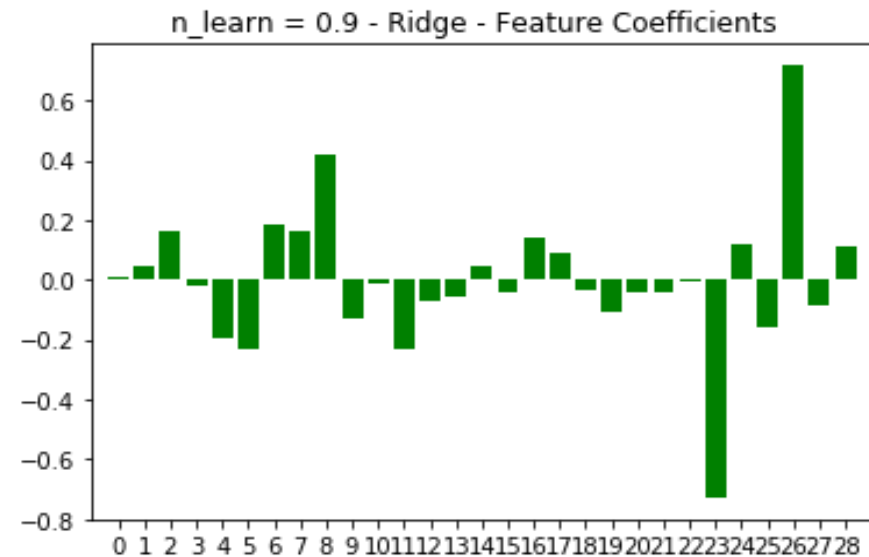
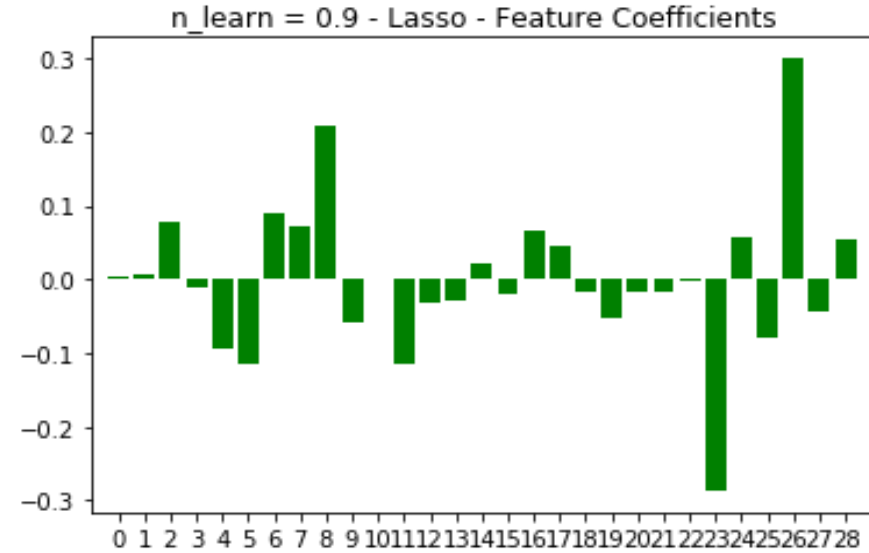
MODEL	TIME (SECONDS)	MINI ERROR RATE	N-LEARN	COMMENTS
RIDGE	0.10	0.14	0.5	
RIDGE	0.11	0.14	0.9	
SVM	0.32	0.11	0.5	TIME increase a lot when n-learn increase
LASSO	0.46	0.24	0.5	
LASSO	0.48	0.24	0.9	
LS	0.76	0.10	0.9	Best performance with efficiency
LS	0.76	0.09	0.5	
SVM	0.96	0.11	0.9	TIME increase a lot when n-learn increase
RF	6.29	0.08	0.5	
RF	10.88	0.10	0.9	Best performance but low efficiency

- FEATURE SELECTION

n-learn = 0.5



n-learn = 0.9



- **INSIGHT**

Important Features for ETF:

- **fund_yield**
- **portfolio_stocks**
- **price_earnings**
- **category_return_1month**
- **net_annual_expense_ratio_fund**
- **net_annual_expense_ratio_category**
- **price_cashflow**
- **rating_aaa**

LS will be the best model for balancing the time and performance

RF will perform the best predictive performance